Predicting Singapore HDB Rents with Comparable Transactions and Temporally Diverse Ensembles

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***Abstract*—We study monthly rental prediction for Singapore Housing and Development Board flats using 150,000 labelled transactions and supplied spatial and economic data. The task combines heterogeneous housing characteristics, a changing market and incomplete identification: identical recorded attributes need not identify the same flat. We develop a pipeline that normalises categories, joins block and amenity information, and summarises strictly earlier comparable rents. The selected predictor averages eight CatBoost components across direct and market-residual targets, two random seeds, and normal versus deliberately stale training histories. It achieves RMSE 440.76 and MAE 316.55 SGD on January–March 2025, compared with RMSE 445.52 for a four-component adaptive-market reference. The selected ensemble also improves recent annual and sixteen-month validation, but worsens an older annual window by 9.81 RMSE. Error analysis shows that transactions at or below 2,500 SGD represent 16.34% of recent validation yet contribute 52.23% of squared error. Bounded experiments with specialists, stacking, cross-fitted statistics, residual corrections, economic indices and MRT opening dates do not establish a reliable replacement. These findings demonstrate the usefulness of comparable-market information while exposing temporal sensitivity and unobserved transaction conditions that aggregate prediction scores alone can conceal.**

***Index Terms*—HDB rental prediction, tabular regression, comparable transactions, temporal validation, ensemble learning**

I. Introduction

Predicting a plausible monthly rent can help renters compare options and help owners assess an asking price. In Singapore's public housing market, a prediction must account for the physical flat and its surroundings, while recognising that the same nominal amount has different market meaning at different dates. A large flat far from transport and a smaller flat near a commercial centre may attract similar rents. We aim to estimate recorded monthly rent from observable transaction characteristics and to explain where those estimates remain unreliable.

The course provides a regression task built from HDB rental approval information and auxiliary datasets [1]. We treat it as dated cross-sectional transaction prediction. The underlying process may contain repeated observations of individual flats, but the released fields do not establish those identities. Several flats in a block can share their type, model and area. A block-level rental history is therefore evidence about comparable transactions, not a verified trajectory for one flat. This distinction matters both when handling repeated records and when interpreting historical features.

Three research questions organise the study. *RQ1:* What information is recovered through consistent preprocessing and spatial joins? *RQ2:* Do market-adjusted comparable rents and different training-history views improve local prediction? *RQ3:* Do these improvements transfer across periods, and which groups account for the remaining errors? Local chronological evaluation and interpretation are our primary evidence; competition scores provide a supplementary external check.

Our contribution is an empirical workflow rather than a new boosting algorithm. We document (1) an auditable treatment of categories, spatial joins and ambiguous repeats; (2) a reproducible predictor combining comparable-market information with training-history diversity; and (3) a temporal and subgroup evaluation that retains adverse findings. The investigation shows why a recent improvement need not imply robustness under a different market regime. Code and cumulative experimental records are maintained in the team repository.[1]

II. Data Understanding and Preparation

A. Released files and task boundaries

The actual training file contains 150,000 rows from January 2021 through March 2025; the unlabelled competition file contains 50,000 rows from March 2025 through July 2026. Ten input fields describe approval month, town, block, street, flat type, flat model, floor area, furnishing status, lease commencement year and fee. Training additionally supplies monthly rent. These file-level counts and dates differ from the course webpage's broader dataset overview; all experiments use the provided files rather than inferring coverage from that overview.

Month bins overlap in March 2025: 3,032 training rows and 380 competition rows occur in that month. Thus the competition period is predominantly later, but month precision does not establish strict chronological separation for every transaction. We do not access hidden test rents. Approval month is not necessarily the month in which rent was negotiated, limiting interpretations of short-term market movement.

Training rent has mean 2,719.04, median 2,700, standard deviation 749.73, and range 300–7,600 SGD. No core input is

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[1] https://github.com/ElephantLikeCode/cs5228-hdb-rental-prediction

missing, although complete fields do not establish correctness or complete economic information. Auxiliary tables describe blocks, MRT stations, malls, primary schools, Certificate of Entitlement (COE) prices and stock prices. A normalised block lookup covers every released rental row without missing coordinates or principal block attributes.

B. Cleaning guided by exploratory analysis

We lowercase categories, trim whitespace, replace punctuation with spaces and collapse repeated spaces. For example, `3-room` and `3 room` become the same category. Raw flat-type labels reduce from eleven to six and street labels from 1,146 to 576 in training. Identical deterministic rules apply to training and competition inputs. This prevents formatting variants from fragmenting comparable groups while preserving immutable raw CSVs.

Furnishing status and fee are constant and are dropped. Calendar year, month, quarter, a month index and nonnegative lease age are derived. Lease age uses the recorded lease commencement year; it is not renovation condition or tenancy duration. Address combines normalised block and street. Block joins also use town and enforce many-to-one matching, preventing accidental row multiplication.

CatBoost receives canonical strings explicitly identified as categorical columns. Linear and Ridge baselines use training-fitted one-hot encoding with unknown values ignored, median imputation and numeric scaling. The scikit-learn tree baselines use training-fitted ordinal encoding, an unknown-category sentinel and numeric median imputation. XGBoost uses native categorical splits with vocabularies fitted on training only. These representations and model capacities differ, so the baseline comparison does not isolate algorithm choice alone.

Fig. 1 motivates two subsequent choices. Monthly medians change markedly, making time-invariant group averages inadequate as the only predictor. Flat-type differences also persist, motivating market normalisation within flat type rather than pooling all room counts into one index. Neither descriptive pattern identifies a causal effect, because transaction composition changes over time.

C. Repeated inputs and extreme rents

An initial audit finds 425 exactly repeated labelled rows. Equal recorded fields and rent are not sufficient evidence of the same flat or contract, so these observations remain. Normalised feature vectors can also repeat with substantially different rents. Removing such records would change the frequency of transaction types without resolving absent individual-flat identity.

Low rents for larger flats motivate a sensitivity study. A fixed rule flags three-room-or-larger or executive observations below half the median for the same approval month, town and canonical flat type. It flags 954 training rows, or 0.636%. Such a target-defined flag is a diagnostic or training-policy intervention; it cannot be computed from an unlabelled test transaction. We evaluate alternative training policies on the same complete local validation population and preserve every competition row in its original order. We neither label all low rents as errors nor multiply them by a presumed room count.

III. Methods

A. Spatial information and baseline learners

Block features include coordinates, maximum floor, completion year, subzone, planning area and region. Great-circle distances connect each block to the nearest MRT station, mall and primary school; counts within 0.5, 1 and 2 km describe nearby amenities. These are geographic proxies rather than walking or transit travel times. Predictive association with proximity must not be described as a causal premium.

Global and town/flat-type medians establish simple references. Linear Regression, Ridge, Decision Tree, Random Forest and histogram gradient boosting span increasingly flexible approaches. Random Forest [2] supplies a nonlinear bagged-tree reference. CatBoost [3] handles the mixture of numeric and high-cardinality categorical variables and becomes our selected learner after empirical comparison. Later, XGBoost [4] receives the same constructed inputs to test learner complementarity. These choices do not imply that CatBoost is universally best.

B. Past-only comparable statistics

For query month t, historical features use only permitted training transactions strictly earlier than t. No query-month rent, including a training row's own rent, enters its statistics. For group g, an initial smoothed mean is

$$\widehat{\mu}_{g,t} = \frac{\sum_{i \in H_{g,t}} y_i + \lambda \mu_t}{|H_{g,t}| + \lambda}, \quad \lambda = 20, \tag{1}$$

where $H_{g,t}$ is the earlier group history and μ_t the earlier global mean. Log support counts accompany these means. Groups include town/type, address/type, address/type/rounded area, street/type and town/type/model, as well as broader groups. These keys identify comparables, not unique flats. Empty early history remains missing with explicit support information; adaptive levels use a fixed 2,000 SGD cold-start prior.

Unadjusted historical rents confound local premium with market movement. Our adaptive representation divides each reference rent by its own flat-type/month mean. It estimates decayed premiums and shrinks specific groups toward parents: town/type, town/street/type, address/type, and address/type/area. Weights $w_i = 2^{(m_i - a)/12}$ are anchored at the latest permitted reference month a, giving

$$\widehat{r}_{g,t} = \frac{\sum_{i \in H_{g,t}} w_i r_i + \alpha_g \widehat{r}_{p(g),t}}{\sum_{i \in H_{g,t}} w_i + \alpha_g}. \tag{2}$$

Here r_i is the reference rent ratio, the top-level parent premium is one, and the four shrinkage strengths are 30, 12, 8 and 5. Anchoring weights to the latest observation avoids making effective support vanish merely because the query is distant. Each group supplies its market-adjusted level, premium, effective support and observation age.

For each flat type, the market forecast begins with the count-weighted mean of the latest three observed monthly means. A

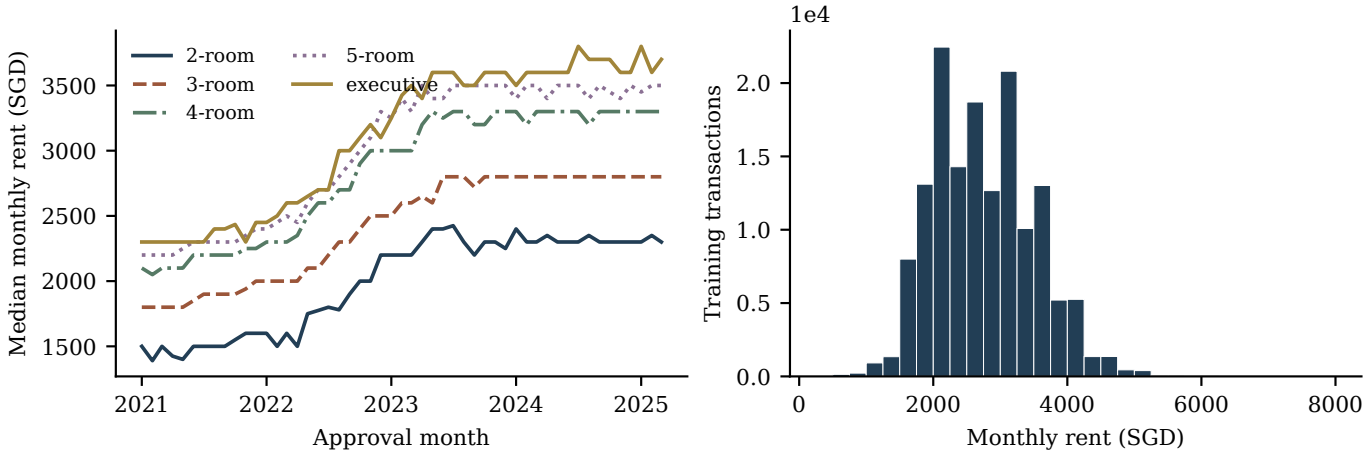


Fig. 1. Training-data profile. Left: monthly median rent by canonical flat type; one-room flats are omitted because there are only 42 training observations. Right: the distribution of all 150,000 training rents. Temporal curves describe changing transaction samples, not repeat-flat or causal price indices.

log-linear slope uses up to twelve months, requires at least six, and is clipped to $[-0.01, 0.01]$ per month. For horizon h from the final reference month,

$$\widehat{M}_{f,t} = M_{f,a} \exp\left[\beta_f \, 6(1 - e^{-h/6})\right]. \tag{3}$$

Multiplying this forecast by (2) gives a comparable-market level. Damping limits extrapolation but does not eliminate regime risk. These coefficients are recorded design choices rather than parameters fitted to outer validation labels.

C. Local trends and training-history diversity

Four town/flat-type features capture movement relative to the flat-type market: normalised premium in the latest six reference months, premium in the preceding six, their log ratio clipped to $[-0.15, 0.15]$, and log support based on the smaller period count. Both period ratios shrink toward one with strength 20. Thus the method measures local relative momentum rather than fitting an unconstrained future trend independently for each block.

A second training view simulates stale information. Eligible observations are copied with the latest six months of reference labels hidden, while preserving their actual query dates. Eligibility requires at least twelve months since the first reference month. The original and copy each receive half the observation's original weight, preserving total weight. This creates alternative information conditions, not new independent rental evidence. At inference, all components use the full permitted training history for each query.

Each view has a direct rent model f and a model g for residual $y - b_v(x)$, where $b_v(x)$ is the adaptive address/type/area level. Averaging both target modes over seeds 42 and 2026 yields

$$\widehat{y}(x) = \frac{1}{8} \sum_{v} \sum_{s} \left[f_{v,s}(x) + b_v(x) + g_{v,s}(x) \right], \quad v \in \{\text{normal}, \text{stale}\}, \quad s \in \{42, 2026\}. \tag{4}$$

The normal view includes local trends; the stale view uses hierarchical features without those four trend columns. Component input dimensions are 78 and 74 respectively. Equal weights preserve diversity without fitting another layer to the latest evaluation errors.

D. Alternative hypotheses

Stacking [5] uses predictions generated by earlier training origins to fit a second layer. We compare four seed-averaged heads with all eight individual outputs, using constrained weighting or Ridge correction. Meta-training always precedes meta-evaluation. The retained probe fits nonnegative sum-to-one four-head weights on October–December 2024 and shrinks them halfway toward equal weights; January–March 2025 labels are not used to fit those weights.

Further bounded screens test nearest comparables, flat-type specialists, probabilistically routed rent-band experts, five-fold cross-fitted statistics, honest out-of-fold (OOF) residual corrections and matched-input learners. OOF predictions come from models not fitted on the predicted row's label. Corrections learn earlier-origin OOF residuals rather than fitted-training residuals. Validation routing uses observed features and predicted probabilities, never actual query rent. An isolated area/age composition-trend ablation preserves the original history levels and residual offset; an earlier joint change is not treated as an isolated test.

Economic screens use lagged supplied COE and stock summaries, then official CPI and URA private non-landed rental indices. CPI measures consumer prices, while URA measures private housing rather than HDB rents. Release-lagged levels and changes are supplied with missingness and staleness indicators. A separate MRT calendar reconstructs operational stations by rental month and deduplicates interchange names. Opening exposure describes distance-decayed new-line access, recent openings and months since opening. These features test predictive association, not a causal opening effect.

IV. Experimental Design

A. Local validation and metrics

Chronological validation reflects the predominantly later competition period without assuming a pure time series. Each local origin fits on earlier months and retains the complete scoring population. The recent quarter uses 141,076 training and 8,924 validation rows. Table II adds annual and sixteen-month windows. These windows overlap and have been repeatedly inspected during development; they are not independent replications or untouched final tests.

RMSE emphasises large deviations, MAE measures typical absolute error, and signed bias distinguishes systematic overprediction from underprediction. Error is prediction minus observation. For subgroup g, squared-error share is $\sum_{i \in g}(\widehat{y}_i - y_i)^2 / \sum_i (\widehat{y}_i - y_i)^2$. Counts accompany subgroup scores: high RMSE in a rare group can contribute less total error than moderate RMSE in a common group.

Historical exploration used scoring-window early stopping for CatBoost, with patience 100. Those results are descriptive references, not unbiased final comparisons. Subsequent temporal comparisons fix 1,600 trees per component and supply no outer scoring labels for early stopping. We keep the protocols separate: a changed tree-selection procedure must not be presented as a gain caused solely by a new feature. Neither procedure uses hidden competition labels.

B. Budgets, promotion and uncertainty

Selected CatBoost settings are learning rate 0.04, depth 8 and leaf L_2 regularisation 6. Early recorded alternatives include 1,200 and 1,600 trees at learning rate 0.04, and 2,200 at 0.03 with regularisation 8. We subsequently freeze 1,600 trees so deployment and local comparison use the same fitting budget. Depth-6 models, recent-sample weighting and market-normalised residual targets are bounded checks rather than an exhaustive parameter search.

Later research usually starts with seed 42 on the recent quarter and sixteen-month window. Candidates replace the matching direct/residual pair in the frozen eight-component mainline, or enter a predefined 25% blend. These are deployment screens, not full eight-component retrainings or equal-compute learner comparisons. A meaningful recent reduction, approximately 2 RMSE, plus a positive long-window gain is required before another seed and further confirmation periods. This gate discourages promoting changes comparable to fitting noise.

Selected paired comparisons resample normalised block-address clusters 1,000 times and recompute RMSE gains, reporting the 2.5th and 97.5th percentiles. These exploratory intervals preserve within-address association but do not remove dependence across months or adjust for multiple candidate searches. They are not uncertainty intervals for future rents or private competition scores. CatBoost GPU reductions can vary across nominally identical fits; saved predictions and source/input hashes support auditability, not bit-identical refitting.

TABLE I
Historical January–March 2025 core-input references. CatBoost uses scoring-window early stopping; this exploratory protocol is separate from the fixed-budget ensemble study. Errors are SGD.

Historical core model	RMSE	MAE
Global median	844.60	691.33
Town/type median	546.81	425.11
Linear regression	523.20	373.09
Ridge	520.01	366.58
Decision tree	485.90	358.59
Random forest	470.19	348.52
Histogram boosting	463.36	336.75
CatBoost (core)	455.71	337.09

V. Results and Interpretation

A. Baseline and spatial information

In Table I, the global median poorly accommodates heterogeneous flats and the changed market. Town/type medians improve markedly, and nonlinear trees improve further under the chosen representations and budgets. Historical spatial ablations give CatBoost core RMSE 455.71, block 454.19, block plus MRT 452.89, block plus malls 453.97, block plus schools 453.71, and all spatial features 452.62. Modest incremental gains do not establish small real-world amenity effects: address categories already encode correlated location information, and snapshots are imperfect proxies.

The detailed-comparable two-seed model reaches 448.06 under historical early stopping. Later fixed-budget adaptive direct/residual averaging gives 445.52 on the same recent rows. These are milestones under different protocols, so their difference is not an isolated feature gain. Table II supplies the controlled fixed-budget evidence for subsequent selection.

B. Recent gains and an older-period regression

History diversity reduces adaptive-reference RMSE by 4.63 on the recent quarter, 2.65 on the recent year and 2.81 on sixteen months, but worsens the older year by 10.55. Adding local trends improves diversity by another 0.13, 0.13, 0.74 and 0.40 respectively. The selected recipe therefore still loses 9.81 versus the adaptive reference on the older year. Its older bias is +82.72 SGD, compared with +8.89 for the adaptive reference. Improvement is conditional on market period.

The selected recent-quarter RMSE is 440.76, MAE 316.55 and bias +0.79 SGD. Near-zero aggregate bias does not imply calibration within each location or price band. Fig. 2 motivates retaining the simpler reference and explicitly acknowledging unresolved temporal sensitivity. Simulating stale history changes the training distribution and can help one regime while overcompensating in another.

C. Stacking and supplementary competition evidence

Four-head recent-window stacking improves local recent RMSE from 440.7629 to 440.6565, but its paired gain interval $[-0.0342, 0.2262]$ crosses zero. The eight-output version instead scores 440.8165. Earlier transfer tests are inconsistent;

TABLE II

LOCAL FIXED-BUDGET TEMPORAL COMPARISON. THE ADAPTIVE REFERENCE HAS FOUR COMPONENTS; HISTORY DIVERSITY AND THE SELECTED LOCAL-TREND ENSEMBLE HAVE EIGHT. EACH COMPONENT USES 1,600 TREES; BOTH SEEDS ARE INCLUDED. LOWER RMSE IS BETTER. WINDOWS OVERLAP.

Validation interval	n	Adaptive (4)	Diversity (8)	Local trend (8)
Jan-Mar 2025	8924	445.52	440.89	440.76
Apr 2024-Mar 2025	35158	469.54	466.89	466.76
Apr 2023-Mar 2024	35882	580.54	591.08	590.35
Dec 2023-Mar 2025	47171	486.64	483.83	483.43

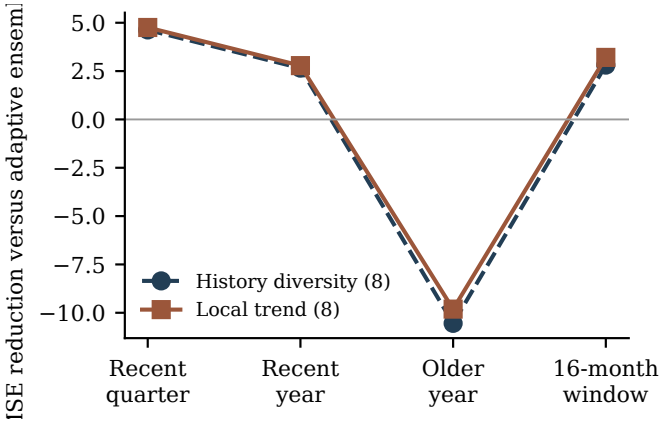


Fig. 2. RMSE reductions relative to the four-component adaptive-market reference. Both diversity designs improve three development windows but regress on the older year. Positive values mean lower error; this is a window comparison, not an uncertainty plot.

a sixteen-month stress check loses 0.538 with four inputs and 1.189 with eight. That check changes regime, fitting interval and horizon together, so it does not isolate recency weighting. Stacking remains an alternative rather than the default.

Public submissions provide supplementary evidence, not the primary model-selection criterion. The ledger records 438.94488 initially, 436.05684 for adaptive averaging, 432.47187 for history diversity, 432.14881 for local trends, and 432.04286 for the stacking probe. The last two were team-reported historical scores, not a fresh leaderboard reading. Their hidden public subset differs from local validation. The stacking public reduction of 0.10595 does not override inconsistent local transfer.

Earlier keep-versus-remove low-rent submissions score 436.75718 and 436.58736, a difference of 0.16982 in that earlier model family. This offers little practical evidence for deletion and does not establish that flagged labels are wrong. These older submissions are distinct from the later ensembles. We retain all labels in the selected model; removing difficult validation observations would merely change the evaluated population.

D. Where errors concentrate

Table III and Fig. 3 show that 1,458 rents at or below 2,500 SGD are 16.34% of recent rows but contribute 52.23% of squared error; the corresponding long-window share is

TABLE III

SELECTED-MAINLINE RECENT VALIDATION ERRORS, ALL 8,924 ROWS. BIAS IS PREDICTION MINUS OBSERVATION. ACTUAL-RENT BANDS ARE DIAGNOSTIC ONLY AND CANNOT ROUTE AN UNLABELLED TRANSACTION.

Actual rent	n	RMSE	Bias	SSE (%)
$\leq$2500	1458	788.1	+589.6	52.2
2501-3000	2525	294.4	+47.3	12.6
3001-3500	2823	276.2	-113.2	12.4
3501-4000	1541	366.7	-252.8	12.0
$>$4000	577	568.9	-456.0	10.8

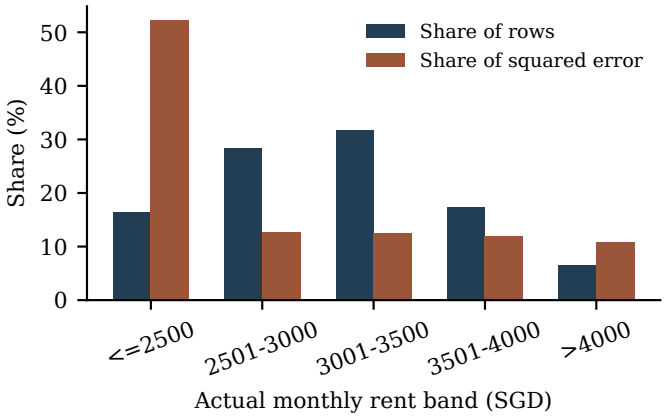


Fig. 3. Share of recent observations versus share of mainline squared error. The lowest actual-rent band dominates squared error disproportionately. Membership uses observed targets and is unavailable at prediction time.

50.86%. This broad band includes plausible low-priced flats and is not an anomaly label. Low observed rents are overpredicted and high rents generally underpredicted, consistent with shrinkage toward typical conditional rents when transaction conditions are unobserved.

Executive flats have recent RMSE 506.73 across 462 rows, yet only 6.84% of squared error. Three- and four-room groups together contribute 64.87%, partly reflecting prevalence. A narrower diagnostic event, actual rent below 65% of the frozen prediction, occurs in 2.992% of recent rows and contributes 32.47% of squared error. A feature-based detector has AUC 0.5761 and average precision 0.04161, versus prevalence 0.02992. This weak ranking signal supports neither reliable hard routing nor a claim that these events are completely unpredictable.

Large errors are therefore not restricted to rare executive

TABLE IV
RESEARCH SCREENS APPLIED TO THE FROZEN MAINLINE; ERRORS ARE SGD. BLEND ROWS USE 25% CANDIDATE WEIGHT, AND OOF CORRECTION USES 25% RESIDUAL STRENGTH. OTHER ROWS REPLACE TWO OF EIGHT COMPONENTS. SCREENS USE ONE SEED AND DIFFERING BUDGETS; NO ROW ESTABLISHES A RELIABLE REPLACEMENT.

Candidate applied to mainline	Recent	16-month
Nearest-comparable blend	441.842	482.198
Flat-type specialist blend	441.261	483.356
Rent-band mixture blend	441.962	482.488
Cross-fitted statistics	442.344	483.672
Strict-past distribution	440.641	483.299
COE indicators	440.877	483.434
Stock indicators	440.816	483.536
Matched-input XGBoost	441.881	485.244
OOF residual correction	440.922	483.340
Guarded Ridge blend	442.398	481.515
CPI	440.860	483.557
URA rental index	440.878	483.528
Corrected static MRT	440.631	483.418
MRT opening exposure	440.760	483.352
Monthly-available CPI+URA	440.791	483.478

flats. Contract terms, condition, renovation, household relationships or label problems are possible explanations for low rents, but none is established by released fields. A target-defined error group helps investigation; it is not itself an operational detector. The next useful analysis should seek a stable observable signal across periods before fitting another specialist.

E. Bounded extensions and negative evidence

Table IV compares distinct hypotheses, not equal-budget algorithms. Nearest-comparable, specialist and soft-mixture blends can improve sixteen months while losing recently. Matched-input XGBoost worsens both windows; highly correlated prediction errors limit the tested blend's complementarity. This supports retaining the present recipe within the tested budget, not rejecting either algorithm or specialist modelling in general.

Cross-fitting tests a cross-sectional encoding without unique-flat assumptions. Training statistics exclude a held-out inner fold; outer validation uses only outer-training labels. Full-interval cross-fitted statistics worsen both windows. Strict-past dispersion replacement gains only about 0.12 and 0.14 RMSE. The encoding changes training/inference support distributions, so these results apply to the tested configuration. An isolated area/age composition-trend ablation also provides essentially no recent gain and worsens the other main windows.

COE and stock features use preceding-month summaries and origin-frozen controls, but a monthly lag is only a coarse release-time proxy and stock membership is a supplied snapshot. Neither gives a material deployment gain. Ridge reveals a different issue: future numeric inputs can exceed training ranges. Joint clipping to training ranges substantially stabilises its predictions, yet its fixed blend still loses recently while gaining over sixteen months. Since several fields change together, this does not identify one causal extrapolation variable.

For official macro indices, month-start queries activate CPI [6] two months after its observation month and URA two months after quarter end. Main screens freeze indices at the validation origin. A same-fitted-model control supplies values released by each rental month, testing contemporaneous covariates rather than a single-origin forecast. Historical revisions and CPI rebasing are not reconstructed: this is latest-vintage, release-lagged evaluation, not fully point-in-time vintages. URA private non-landed rents [7] are only a proxy for HDB conditions. The monthly CPI+URA replacement remains slightly worse than the mainline on both windows despite improving its own long-window frozen-input control.

The MRT audit finds 55 supplied rows marked open outside the verified operational calendar. The selected model still uses the supplied snapshot, an explicit availability limitation. Research variants verify TEL2–4, Punggol Coast and Hume dates against LTA releases (for example, TEL4 [8]), exclude future/unverified codes, deduplicate interchange names, and activate service from the first full month after opening. Origin-based validation admits dated openings announced before that origin; the monthly-access control admits operational service by the rental month. Supplied coordinates limit coverage, especially for LRT. Corrected static rail gains just 0.1322 and 0.0169 RMSE, with intervals crossing zero. Filtering and deduplication change jointly, so their contributions are not isolated. Dynamic access and opening exposure establish neither a reliable improvement nor a causal premium.

The index and rail studies comprise 36 formal GPU fits, excluding smoke checks. Repeating an otherwise identical seed-42 macro fit changes a replacement score by about 0.13 despite identical input matrices. Improvements at this scale require caution. None passes the advancement gate, so rejected screens do not produce a new final model or submission.

VI. LIMITATIONS AND REPRODUCIBILITY

A. Scope of the conclusions

Evaluation reuse. Temporal windows have been inspected repeatedly, several overlap, and public submissions add another selection channel. Bootstrap intervals do not repair this reuse. We have no private-label evaluation. The older-period regression prevents a claim of general dominance over the adaptive reference; leaderboard scores cannot reveal how other teams obtained much lower error.

Incomplete observation. Matching input attributes need not identify one flat or contract. Deleting equal rows or treating every extreme as an error would assume unavailable information. Conversely, ambiguity does not establish a population irreducible-error floor. Label-informed oracle diagnostics are not deployable predictions. Unit floor, renovation, precise walking access and contract terms remain unobserved. Predictions estimate recorded transactions, not a normative fair rent.

Information availability. Mainline MRT fields are static proxies; reconstructed dates improve semantics but the tested features do not provide strong score evidence. CPI/URA

historical vintages and stock membership are not fully reconstructed, and approval can lag negotiation. These limitations distinguish offline competition prediction from a live forecasting service. No spatial importance or opening association is a causal estimate.

Bounded search. Parameters and representations are not exhaustively explored. One-seed, two-component screens cannot rule out better full retraining. No uncertainty interval for an individual tenancy is supplied. Future work should first identify stable observable subgroup bias in additional, preferably non-overlapping periods, then confirm a promising correction on later data and repeated seeds. Walking access, network connections or commuting time offer transit hypotheses more specific than repeated threshold tuning. These are proposals, not completed experiments.

B. Implementation and reproducibility

Reusable code lives in `src/`; `configs/model.json` records the selected recipe. The project maintains one cumulative progress record, score ledgers, model metadata and reproduction commands. Raw data is immutable. Local prediction arrays and source snapshots permit metric verification; data, models, competition CSVs and local AI notes are excluded from GitHub. A fresh reader obtains the course data through the authorised channel and regenerates caches using the README.

The recorded eight-component full-data training took 669.44 seconds on an RTX 4060. This measures one run, not total project computation. Fit timers must not be summed over multiple rows reusing the same fit or reference. The environment records Python 3.12 and pinned dependencies, including CatBoost 1.2.10. The 6 October implementation passes 55 automated tests covering feature cutoffs, canonical categories, joins, output protection and availability controls. Report figures and tables are regenerated from existing evidence by `final_report/prepare_report.py`, without model training. Appendix A maps the supporting evidence.

VII. Conclusion

Comparable-market information and training-history diversity improve recent HDB rental prediction, but benefits remain conditional on period and information availability. The selected ensemble reaches local RMSE 440.76 and MAE 316.55 on the recent quarter, while an older annual deficit of 9.81 versus the adaptive reference remains unresolved. Low observed rents dominate squared error disproportionately, yet available features provide only weak separation of the most unusual cases. Bounded negative experiments delimit the claims supported by the present data. A defensible system therefore needs explicit data assumptions, comparable validation and interpretable failures alongside its prediction score.

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APPENDIX A
REPRODUCTION AND EVIDENCE MAP

Obtain the original course files under `data/raw/`, then run `python src/check_setup.py` and `python -m pytest -q`. The README supplies full commands to regenerate temporal component predictions before feedback studies. Cached predictions and models are intentionally not distributed with the report.

Fixed-budget comparisons use `src/run_experiments.py` with the `--compare-candidates` option. The selected ensemble is generated by `src/train_final_model.py` with the recorded configuration; research uses `src/feedback_experiments.py`, including `--families public` for indices and rail. These are abbreviated entry points; use the complete README commands for exact windows and options.

For clarity, the historical core-baseline settings are listed in Table V. They are bounded references, not the outcome of an exhaustive or common-compute search. The selected ensemble's settings and history coefficients are given in the main text. The IEEEtran class is unchanged from the official Canvas template. Figures and tables derive from the team's own records. Earlier public CS5228 reports informed presentation structure, not experimental results or this edition's requirements. Public scores are historical ledger entries, not current-rank claims.

TABLE V
PRINCIPAL HISTORICAL BASELINE SETTINGS; PREPROCESSING IS DESCRIBED IN THE MAIN TEXT.

Model	Recorded settings
Linear/Ridge	Ordinary least squares; Ridge $\alpha = 10$
Decision Tree	Maximum depth 18; minimum leaf size 20
Random Forest	180 trees; maximum feature fraction 0.7; minimum leaf size 3
Histogram boosting	250 iterations; learning rate 0.08; 31 leaves; $L_2 = 2$

TABLE VI
EVIDENCE LOCATIONS; DETAILED LOCAL PREDICTIONS CAN BE REGENERATED.

Evidence	Location
Baseline/spatial results	`reports/experiment_results.csv`
Fixed-budget windows	`reports/temporal_summary.csv`
Research screens	`reports/feedback_results.csv`
Stacking transfer	`reports/stacking_results.csv`
Actual public scores	`reports/kaggle_submissions.csv`
Selected recipe	`configs/model.json`
Model provenance	`reports/final_model_metadata.json`
Decisions/limitations	`docs/PROGRESS.md`

APPENDIX B
TEAM WORKLOAD STATEMENT – TO BE COMPLETED

Reserved for completion before final submission. Responsibilities and contribution proportions have not been assigned in this draft.

TABLE VII
WORKLOAD PLACEHOLDER; NO CONTRIBUTIONS ARE ASSERTED.

Member	Student ID	Responsibilities
Han Linxuan	A0350911M	To be completed
Sun Shenghe	A0354849L	To be completed
Zhang Yuhong	A0356795H	To be completed
Zhang Zhexuan	A0355290A	To be completed

APPENDIX C
AI ASSISTANCE AND DRAFT STATUS

AI tools assisted with discussion of experimental ideas, code development and checks, interpretation of recorded results, and drafting and editing this report. Numerical results derive from executed experiments and saved evidence; proposed analyses are identified separately. The course policy permits AI use for the group project with appropriate documentation.[2] The team must review the report, validate interpretations and complete workload attribution before submission; this statement does not assert that review has occurred. The final deadline remains unconfirmed on the project page checked on 23 October 2026.

[2]https://canvas.nus.edu.sg/courses/93866/pages/policies